

BOL'SHIYE UKI, Russian Federation

WMO: 284910

Lat: 56.933N

Lon: 72.667E

Elev: 285

StdP: 14.54

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-26.0	-36.1	0.7	-31.2	-30.8	1.0	-25.8	14.6	12.8	13.4	11.9	1.9	170	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	83.5	68.3	80.7	66.2	77.8	65.0	70.0	80.8	68.1	78.1	66.2	75.5	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	97.0	75.7	64.2	90.8	73.6	62.5	85.4	71.4	34.2	81.2	32.6	78.0	31.1	75.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	13.5	11.9	-37.8	89.0	7.0	3.0	-42.9	91.1	-47.0	92.9	-50.9	94.5	-56.0	96.7
			-37.9	72.9	6.9	2.8	-42.9	74.9	-47.0	76.5	-50.9	78.1	-56.0	80.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	-0.2	5.2	20.9	37.2	51.0	61.1	64.3	59.2	48.8	36.5	18.5	6.1
	DBStd	24.84	14.49	13.39	10.48	10.21	9.81	8.26	6.62	6.92	7.90	8.93	13.86	14.71
	HDD50	7089	1558	1255	903	397	111	12	0	7	118	424	944	1360
	HDD65	11404	2023	1675	1368	833	443	167	97	206	489	884	1394	1825
	CDD50	1323	0	0	0	14	142	346	443	292	81	5	0	0
	CDD65	164	0	0	0	0	9	52	75	26	2	0	0	0
	CDH74	1578	0	0	0	3	171	511	643	225	25	0	0	0
	CDH80	314	0	0	0	0	37	112	122	41	2	0	0	0
Wind	WSAvg	5.3	4.9	5.1	5.9	6.3	6.2	4.9	4.6	4.5	5.0	5.6	5.7	5.5
Precipitation	PrecAvg	19.4	0.9	0.7	0.9	1.1	1.6	2.7	2.9	2.7	1.9	1.5	1.4	1.3
	PrecMax	26.2	1.8	1.6	2.0	2.5	4.3	5.7	6.1	4.3	4.0	3.4	2.5	2.4
	PrecMin	13.0	0.1	0.0	0.0	0.1	0.4	0.9	1.1	1.1	0.3	0.2	0.3	0.5
	PrecStd	2.9	0.5	0.3	0.6	0.6	0.9	1.3	1.2	0.9	0.9	0.7	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	35.6	46.2	70.7	84.7	87.1	86.8	84.6	77.4	62.8	43.5	35.2
		MCWB	30.4	32.2	37.6	55.5	64.6	68.5	71.7	70.0	61.5	50.8	40.1	33.8
	2%	DB	28.5	31.7	40.7	63.5	78.5	83.4	83.3	79.6	71.0	58.1	39.6	32.5
		MCWB	27.3	29.5	35.0	50.8	61.3	67.6	69.1	66.5	58.2	48.7	37.4	30.8
	5%	DB	24.8	28.0	37.2	58.0	74.1	80.1	80.6	76.0	66.0	53.4	37.1	29.1
		MCWB	23.6	26.3	32.9	47.6	58.6	65.9	66.9	64.6	56.0	46.4	35.1	27.7
	10%	DB	20.7	24.0	34.7	53.4	69.1	76.5	77.8	72.2	62.0	49.5	34.3	25.5
		MCWB	19.6	22.2	31.4	44.8	55.7	64.2	66.1	62.8	53.6	44.3	32.7	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	32.8	38.5	55.8	66.0	72.0	72.9	70.9	63.2	53.0	40.8	34.2
		MCDB	31.5	35.3	44.6	69.6	81.7	82.9	83.9	81.3	73.7	59.3	42.7	35.2
	2%	WB	27.4	29.4	35.6	51.5	62.1	69.0	70.6	68.0	59.5	49.8	37.8	30.8
		MCDB	28.5	31.5	39.5	62.0	76.7	80.1	81.2	77.1	68.6	56.2	39.4	32.2
	5%	WB	23.7	26.3	33.6	48.2	59.6	67.1	68.8	65.7	57.0	47.1	35.2	27.8
		MCDB	24.8	28.0	36.7	57.1	72.2	78.0	78.4	73.7	64.7	52.3	36.8	29.2
	10%	WB	19.7	22.4	31.6	45.1	56.7	65.2	66.9	63.7	54.3	44.5	32.9	24.2
		MCDB	20.7	23.9	34.7	52.6	67.3	74.8	75.7	70.8	61.0	49.5	34.2	25.4
Mean Daily Temperature Range	5% DB	MDBR	13.9	17.0	16.8	16.9	20.5	18.7	17.7	16.9	15.8	12.1	11.0	12.9
		MCDBR	12.5	13.6	16.0	24.1	28.1	24.6	22.3	23.6	23.3	19.5	9.6	11.6
		MCWBR	12.1	12.4	11.9	14.6	14.4	12.1	10.5	11.9	12.9	12.6	8.6	11.2
	5% WB	MCDBR	12.3	13.3	13.9	22.9	25.9	22.3	20.4	20.3	21.3	17.7	9.3	11.6
		MCWBR	12.0	12.3	11.1	14.5	14.6	11.6	10.5	11.0	12.7	12.5	8.8	11.4
Clear-Sky Solar Irradiance	taub	0.275	0.311	0.368	0.413	0.405	0.408	0.426	0.424	0.372	0.351	0.309	0.279	
	taud	2.030	2.015	1.928	2.039	2.228	2.287	2.235	2.214	2.322	2.220	2.077	1.971	
	Ebn at Noon	180	224	239	251	263	264	255	246	244	210	162	137	
	Edh at Noon	19	30	43	46	40	39	40	38	30	25	19	16	
All-Sky Solar Radiation	RadAvg	238	563	1033	1428	1656	1799	1682	1312	864	431	198	144	
	RadStd	21	37	72	116	140	143	132	119	92	51	36	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page